VAISHNAV RAJA

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Experience

Philips — **Cambridge, MA**
R&D Engineering Intern, Surgical Robotics — ***May 2024 – Dec 2024***

- Built a Python automation and evaluation pipeline for a proprietary multi-arm robot to quantify feasible task envelopes and aggregate validation KPIs (reachability/manipulability, singularity margins, collision/occupied-volume checks) into scorecards for design reviews.
- Designed parameterized robot experiments across 8 base layouts under real constraints (coaxial alignment, buffers, joint limits, human-safe zones); automated data collection/processing into comparable visuals that enabled a down-select to 3 configurations.
- Implemented analytical + Jacobian DLS IK with joint-limit handling; used kinematics-based metrics to validate candidate task poses/waypoints and diagnose near-singular behavior.
- Refactored the codebase 700 $\rightarrow$ 300 LOC ($\sim$57%) into a reusable in-house tool with stable configs/IO/logging and test-driven checks; accelerated sweeps via multiprocessing and Numba.
- Prototyped tool-controller variants and supported initial EtherCAT comms bring-up, integration, and early system debug across distributed controllers.

Tata Consultancy Services – Stellantis — **Chennai, India**
Component Design Engineer, Infotainment Systems & Accessories — ***May 2021 – Aug 2023***

- Led end-to-end design & release of a shark-fin antenna; delivered a single global-compliant design reusable across 4 regions (IND/ASEAN/DMOA/LATAM).
- Owned late-phase validation/updates for 8 IVI components (antennas, USB, speakers, RVC); coordinated suppliers and ENOVIA change control through SOP.
- Executed CAD/PCB checks (CATIA) and CAE (Simulia); released drawings/BoMs and validation docs via ENOVIA PLM.
- Eliminated DRL-induced speaker noise and alternator whine via EMI/EMC filters; calibrated IVI audio for 4 regions to clear acoustic DV/PV and OEM NVH targets.
- Maintained DVP&R, DFMEA, and PPAP/APQP artifacts; supported diagnostics/OBD investigations and cross-functional sign-offs with manufacturing/production/QA and suppliers.

Projects

GNSS-aided 3D LiDAR SLAM (Lego-LOAM + GPS)

- Integrated an absolute GPS unary factor into a factor-graph back-end (GTSAM iSAM2) to reduce long-horizon drift in a Lego-LOAM pipeline; exposed localization/mapping outputs through ROS2 interfaces for downstream robotics stacks.
- Implemented LLA$\rightarrow$local conversion (GeographicLib), adaptive noise/gating, and evaluated on UTBM/Mulran to improve robustness on mobile robot trajectories.

Industrial Systems – AGV + iWarehouse + Intelligent Stock Mgmt

- Built a three-wheeled AGV (two powered steppers + caster): Raspberry Pi controller, two Arduinos running per-wheel PID, and a custom motor driver; tuned smooth docking and predictable task execution.
- Integrated Pick-to-Light with weight-based bin sensing and IoT stock monitoring (ESP8266 + load cells, smoothing/thresholding); edge-to-server telemetry via local Pi broker; received Accenture Best Innovator Award (2019).

Learning-Based Feature Selection for ORB-SLAM3 (YOLO Filtering)

- Integrated YOLOv5 detections with ORB-SLAM3 (C++) to remove ORB features inside dynamic-object boxes (person/car/truck), keeping tracking/mapping/loop-closure unchanged.
- Benchmarked EuRoC and KITTI 00 ($\sim$0.5% drift parity) and a custom driving dataset; improved tracking when dynamic features dominated.

Autonomous Robot Navigation with Deep RL (DDPG/TD3)

- Trained a differential-drive robot in a custom 2D EscapeRoom simulator using DDPG and TD3; continuous actions (L/R wheel speeds), shaped reward balancing progress, heading, collisions, and success bonus.
- Performed training/evaluation analysis; TD3 improved stability vs DDPG via double critics, delayed policy updates, and target smoothing.

Full list of projects available at: vaishnavraja11.github.io/portfolio

Technical Skills

Programming: Python, C/C++, Bash/Shell, SQL.
Robotics/Controls: ROS2/DDS, MoveIt, MATLAB/Simulink, kinematics/IK, Jacobian methods, Gazebo, Isaac Sim, MuJoCo, EtherCAT; UR5, Kinova Arms, Mitsubishi MELFA, KUKA KRC.
ML/Vision: PyTorch, OpenCV, YOLO (Ultralytics), pose estimation, multi-view geometry, Visual SLAM, LiDAR SLAM, GTSAM/iSAM2, ORB-SLAM3, sensor fusion, deep reinforcement learning (DDPG, TD3).
CAD/CAE: CATIA, SolidWorks, Simulia, Ansys.
Tooling/Dev Tools: Git, Linux, CMake, GitLab CI/CD, unit/regression testing (TDD), ENOVIA PLM, Mitsubishi GX Works, Rhino/Grasshopper.
Hardware: Raspberry Pi, Arduino, ESP modules, KiCad, SPICE, 3D printing, CNC/machining.

Education

Northeastern University — **Boston, MA**
Master of Science in Robotics — ***Aug 2025***

- Relevant Courses: Mobile Robotics, Control Systems, Sensors & Navigation, Reinforcement Learning, Computer Vision, Algorithms.

Anna University — **Chennai, India**
Bachelor of Engineering in Mechatronics Engineering — ***Apr 2021***

- Relevant Courses: Industrial Automation, Sensors & Signal Processing, Kinematics & Dynamics of Machines, Mechatronics Systems.